ACSNANO

www.acsnano.org

ARTICLE

Artificial Intelligence-Enabled Caregiving Walking Stick Powered by Ultra-Low-Frequency Human Motion

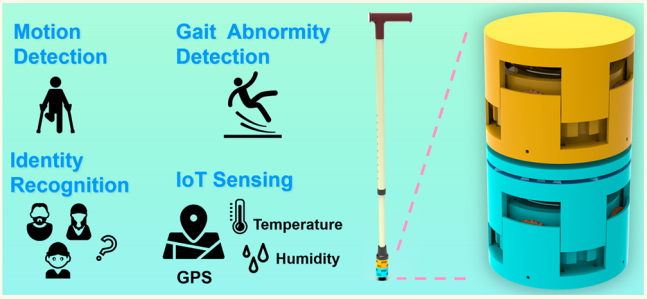
Xinge Guo,[#] Tianyiyi He,[#] Zixuan Zhang,[#] Anxin Luo,[#] Fei Wang,[*] Eldwin J. Ng, Yao Zhu, Huicong Liu, and Chengkuo Lee[*]

Cite This: *ACS Nano* 2021, 15, 19054−19069

Read Online

ACCESS | Metrics & More | Article Recommendations | Supporting Information

ABSTRACT: **The increasing population of the elderly and motion-impaired people brings a huge challenge to our social system. However, the walking stick as their essential tool has rarely been investigated into its potential capabilities beyond basic physical support, such as activity monitoring, tracing, and accident alert. Here, we report a walking stick powered by ultra-low-frequency human motion and equipped with deep-learning-enabled advanced sensing features to provide a healthcare-monitoring platform for motion-impaired users. A linear-to-rotary structure is designed to achieve highly efficient energy harvesting from the linear motion of a walking stick with ultralow frequency. Besides, two kinds of self-powered triboelectric sensors are proposed and integrated to extract the motion features of the walking stick. Augmented sensing functionalities with high accuracies have been enabled by deep-learning-based data analysis, including identity recognition, disability evaluation, and motion status distinguishing. Furthermore, a self-sustainable Internet of Things (IoT) system with global positioning system tracing and environmental temperature and humidity amenity sensing functions is obtained. Combined with the aforementioned functionalities, this walking stick is demonstrated in various usage scenarios as a caregiver for real-time well-being status and activity monitoring. The caregiving walking stick shows the potential of being an intelligent aid for motion-impaired users to help them live life with adequate autonomy and safety.**



KEYWORDS: *triboelectric, Internet of Things, energy harvesting, artificial intelligence, walking stick*

Entering the 21st century, healthcare for aged people and people with disabilities became a global challenge to our social system. According to the data from United Nations and the World Health Organization, the population of people over 60 years of age will double by 2050 and eventually reach around 22% of the world's population. Besides, the number of people with some form of disability has already exceeded one billion, which is about 15% of the world's population, and it is continuously increasing.[1,2] To help them to live physically and cognitively healthy lives is a quest requiring solutions originating from extensive collaboration among multifarious technology domains, *i.e.*, medicine, public health, biotechnology, engineering, and even social science. There has been a flourish of reports on wearable sensors and portable electronics for point-of-care applications, for example, on-body real-time vital sign data monitoring, in the past few decades.[3−8] With the aid of gradually established 5 G and the Internet of Things (IoT) infrastructure, the interaction with the world for elderly and motion-impaired persons has also been increased, resulting in more opportune assistance and rehabilitation.[9−16] Walking sticks, as one of the mobility aids for elderly and motion-impaired people, have been an indispensable part of their daily life by improving their walking pattern, balance, and safety. Beyond these basic functions, together with other well-developed wearable electronics, the walking stick can also become a potential monitoring platform to give users more comprehensive and intelligent help in this era. The synchronous kinematic linkage between the walking stick and the human body is generally observed in assisted ambulation; thus the movement patterns of the walking stick

Received: May 26, 2021
Accepted: July 21, 2021
Published: July 26, 2021



Downloaded from pubs.acs.org/ancac3/article-pdf/15/12/19054/1609523/nn1c04464.pdf by VELLORE INSTITUTE TECHNOLOGY user on 25 October 2026

© 2021 American Chemical Society

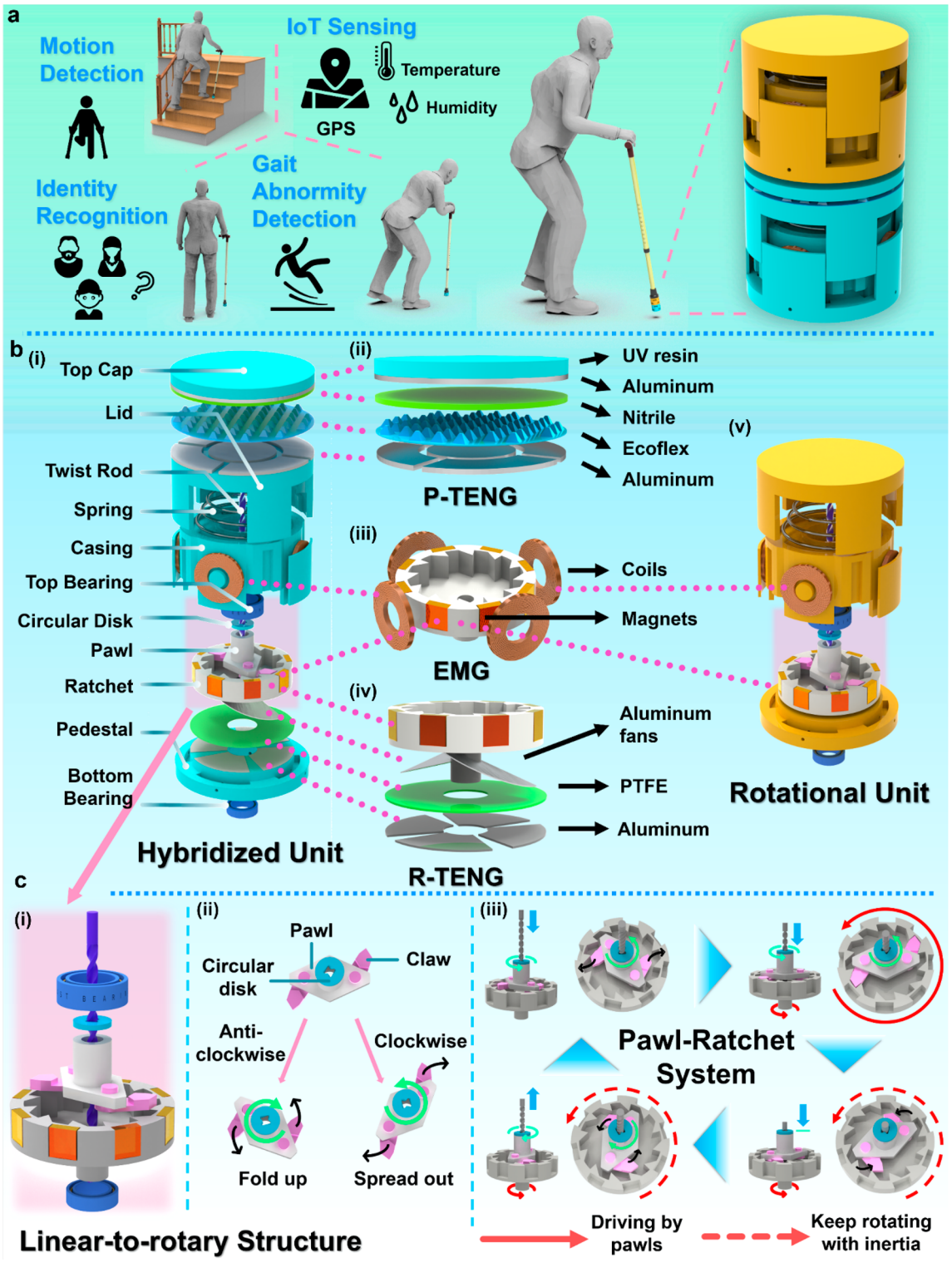


Figure 1. Multifunctional Monitoring system enabled by the caregiving walking stick. (a) Schematics of the caregiving walking stick for realizing multifunctional applications for users. The human model is reproduced with permission from turbosquid.com. (b) Schematics of the hybridized unit and the rotational unit: (i) exploded view of the hybridized unit, (ii) zoom-in schematic of the top P-TENG with corresponding materials for each layer, (iii) zoom-in schematic of the middle EMG consisting of eight magnets and four coils, (iv) zoom-in schematic of the bottom R-TENG with two aluminum fans, (v) exploded view of the rotational unit. (c) Schematics and working principles of the linear-to-rotary structure: (i) exploded view of the linear-to-rotary structure, (ii) working mechanism of the center pawl under two rotation directions, (iii) driving principles of the pawl-ratchet system.

also reflect the activity status of the users. Previous research includes structure innovation as a treatment for Parkinsonian freezing episodes,[17] applying inertial sensors to track users' motion and gait status,[18,19] and equipping distance-sensing functions (*e.g.*, ultrasonic sensor) to help visually impaired users.[20−22] Besides, each strike of the walking stick involves the

https://doi.org/10.1021/acsnano.1c04464
ACS Nano 2021, 15, 19054−19069

Downloaded from pubs.acs.org/ancac3/article-pdf/15/12/19054/1609523/nn1c04464.pdf by VELLORE INSTITUTE TECHNOLOGY user on 25 October 2026

process of transferring weight from the upper limb to the ground, which contains lots of wasted biomechanical energy that can be scavenged for further use.

Aiming at comprehensive monitoring functions, sensors with the ability to fully extract the movement features of a walking stick should be equipped. Currently, microelectromechanical system (MEMS) sensors such as inertial sensors (*i.e.*, gyroscopes and accelerometers), optical sensors, and temperature sensors, are the primary choices for commercialized wearable products such as smart watches, smart phones, and other wearable and portable devices.[23−26] MEMS-based inertial sensors realized by mature silicon MEMS technology ensure high sensing stability and remain as the only commercially viable sensors to measure body motion in our daily life. However, the requirement of an external power supply will also raise the limitation of energy durability, and the abundance of sensing data collected by inertial sensors demands a large computing capacity. On the other hand, self-powered sensors using piezoelectric nanogenerators (PENGs) and/or triboelectric nanogenerators (TENGs) have received growing attention. With self-generated voltage due to mechanical stimuli, they can realize self-powered sensing regarding force, pulse, strain, gas, chemical, *etc.*[27−34] Among them, the TENG stands out for its broad material choices, simple design and fabrication, and low cost.[35−38] Moreover, combining multiple self-powered sensors with artificial intelligence (AI) technology has also become a trend in the development of smart wearable devices, to extract complete sensory information on small-volume data instead of analyzing large-volume data in terms of superficial magnitude and frequency.[39−41]

On the other hand, as the traditional power supply for the aforementioned devices, batteries are limited by their drawbacks including low power density, limited lifespan, and high contamination. At the same time, for elderly or motion-impaired users, the purchase and replacement of batteries for their walking sticks will also become an annoying problem. To either extend the battery life or even replace the battery, energy harvesters (EHs), which can scavenge unused energy from the ambient environment into electricity, have been investigated as a reliable solution.[42,43] Generally, various energy sources are available, such as thermal energy, mechanical energy, and radiation energy, while mechanical energy has excellent energy density with the broadest distribution.[44−46] Previously, EHs were more focused on scavenging mechanical energy from buildings, bridges, and vehicles with a high operating frequency.[47] Nevertheless, biomechanical energy from human motions naturally exists in a low-frequency range around several hertz with more irregular patterns and amplitudes.[48−50] There are two major designs in biomechanical energy harvesters. One is the flexible design mostly based on a stretchable PENG and/or TENG: foldable, durable, washable, and therefore providing wearable comfort.[51−55] The other is the nonflexible design able to further integrate with electromagnetic energy generators (EMGs), which is able to achieve much higher output performance. Concurrently, nonflexible designs may be integrated into wearable platforms, such as watches, bracelets, and accessories on bags and clothes, and realize outstanding power density without compromising wearability.[56−60] Tan *et al.* developed a nonresonant battery-like energy-harvesting module through magnetic levitation.[61] With the hybridization of an EMG, PENG, and TENG, the operational bandwidth from 1 to 5 Hz and a maximum peak power density of 2 mW g^{-1} were achieved. Maharjan *et al.* designed a nonresonant energy harvester based on the spring coupling, with the hybridization of an EMG and TENG, and have realized a working frequency from 0.25 to 6 Hz with a maximum average power of 34.11 mW at 5 Hz.[58] Although the nonresonant structure enables low-frequency energy scavenging, there is still room for improvement in output performance since the maximum output power is proportional to a specific order of the working frequency, which means a higher frequency is more desirable for higher output.[62] Besides, the frequency range shifts to even lower values for elderly and motion-impaired persons, leading to a more enormous challenge for efficient energy harvesting. Accordingly, energy harvesters with frequency upconversion were designed with typical structures like the plucking spring while still operating at tens of hertz.[63,64] Recently, the so-called linear-to-rotary structure was proposed as another method, which is able to transfer linear motion with extremely low frequency to a rotary motion with multiple times higher rotation speed.[65−67] Leveraging this structure, Luo *et al.* proposed a device with an average power density reaching 0.13 mW cm^{-3} at only 0.1 Hz driving frequency.[67]

Combining the self-powered sensors and EHs, a self-sustainable miniaturized system can be realized for broader IoT applications. Aiming for this goal, several recent works have been reported, including a wristband for gas sensing,[68] an armband for electrocardiography,[69] wearable textile for rehabilitation,[70] *etc.* Recently, the artificial intelligence of things (AIoT) has become a burgeoning technology where the IoT system serves akin to a nervous system to deliver essential data collected from multiple sensors to the AI brain for advanced decision making. Diversified and sophisticated applications, such as recognizing human gestures and gaits, can be realized by using TENG sensors through small data volumes with the aid of deep-learning methods.[71−73] Empowered by AI technology, a walking stick can also serve as an intelligent caregiver for real-time monitoring of vital gait, motion, and trace for users and give accident alerts to summon instant medical help. Meanwhile, the walking stick can simultaneously harvest biomechanical energy to enable a self-sustainable IoT system with multiple wireless sensing functions.

In this report, an AI-enabled walking stick powered by ultra-low-frequency human motion is developed to assist the elderly and motion-impaired users in various ways, as shown in Figure 1a. Through the modularly designed units with linear-to-rotary structure, the stick can serve as a reliable solution for users with various requirements, including self-powered sensing and AI-enabled real-time monitoring functions, an ultra-low-frequency energy harvesting function, and a self-sustained IoT sensing function. First, a hybridized unit was designed mainly aiming for sensing-related functions as depicted in Figure 1b(i), which contains a top press TENG (P-TENG) (Figure 1b(ii)), a rotational EMG (Figure 1b(iii)), and a rotational TENG (R-TENG) (Figure 1b(iv)). These three main functional parts can achieve contact point sensing, energy harvesting, and gait abnormality detection, respectively. Moreover, through introducing the deep-learning method to extract high-level features of the sensing signals from the P-TENG, multifunctional real-time monitoring has been successfully achieved, including identity recognition, mobility disability evaluation, and motion detection. Generally, one hybridized unit would be enough to enable such sensing functions of a walking stick. To

Downloaded from pubs.acs.org/ancac3/article-pdf/15/12/19054/1609523/nn1c04464.pdf by VELLORE INSTITUTE TECHNOLOGY user on 25 October 2026

https://doi.org/10.1021/acsnano.1c04464
ACS Nano 2021, 15, 19054−19069

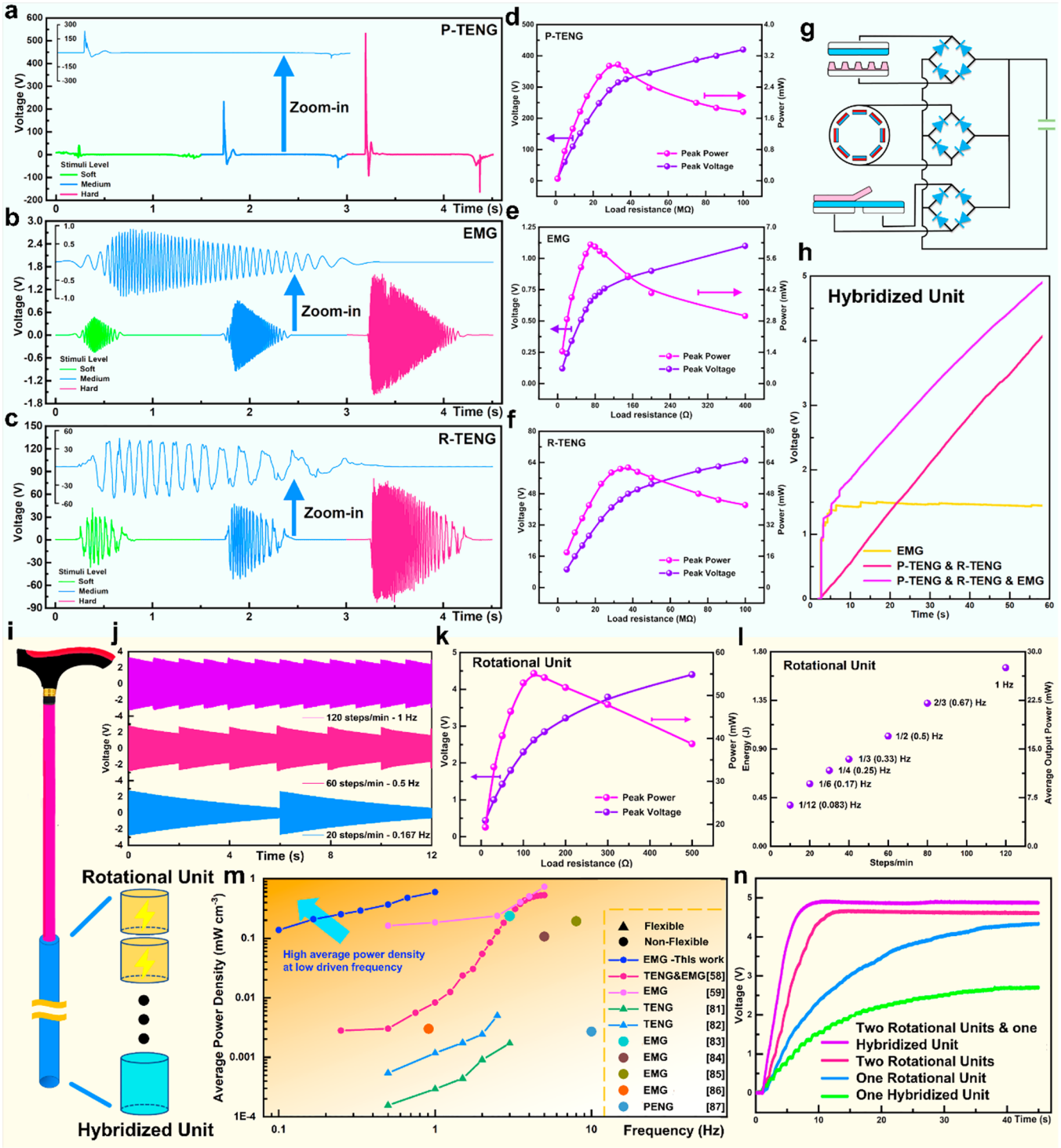


Figure 2. Biomechanical energy harvesting with the hybridized unit and the rotational unit. (a–c) Output voltages under three levels of external stimuli and with a zoom-in output curve under medium stimuli for the (a) P-TENG, (b) EMG, and (c) R-TENG. (d–f) Maximum output voltage and peak output power with the various load resistances for the (d) P-TENG, (e) EMG, and (f) R-TENG. (g) Schematic drawing of the three functional parts for the hybridized unit linked parallelly to charge a capacitor. (h) Charging performance of the hybridized unit for a 10 μF capacitor. (i) Schematic of the modular designed stackable rotational unit aiming at further improving the output performance and fulfilling a more complicated self-sustainable IoT system. (j) Output curve for the rotational unit under various step frequencies measured at the matching resistance within a time interval of 12 s. The walking stick will be pushed every two steps; therefore the driving frequency is half of the step frequency. (k) Maximum output voltage and peak output power of the rotational unit with the various load resistances measured under 1 Hz. (l) Average energy generated in 1 min for a single rotational unit under various step frequencies and corresponding average power. (m) Comparison of the average power density between a rotational unit and previously proposed EHs for scavenging biomechanical energy. For flexible devices, the volume power density is calculated by dividing their area power density by their thickness. (n) Charging performance of stacked rotational units and a hybridized unit for a 4 mF capacitor.

Downloaded from pubs.acs.org/ancac3/article-pdf/15/12/19054/1609523/nn1c04464.pdf by VELLORE INSTITUTE TECHNOLOGY user on 25 October 2026

https://doi.org/10.1021/acsnano.1c04464
ACS Nano 2021, 15, 19054–19069

further leverage the space in a walking stick and scavenge the biomechanical energy from ultra-low-frequency human motion, a rotational unit was proposed (Figure 1b(v)). This unit enables a continuous energy output for a long time interval under one single stimulus and can be simply stacked in the walking stick with the hybridized unit. A rotational unit can provide a peak output power of 55.1 mW. Within 1 min, it can reach an average output power of 27.5 mW at 1 Hz and 6.3 mW at 0.083 Hz, with corresponding average power densities as high as 0.595 and 0.137 mW cm^{-3}, respectively. With two rotational units stacked together for powering, a demonstrative self-sustainable IoT system is presented, equipped with tracing and environment amenity sensing functionalities. Combining these multidimensional sensory features in the caregiving walking stick, comprehensive real-time monitoring reflecting the synchronous location and well-being status of the users becomes feasible, which can provide a timely alarm in the event of sudden accidents.

RESULTS AND DISCUSSION

Key Working Mechanisms Enabling the Caregiving Walking Stick. Although the regular contact and separation motions of the walking stick with the ground are considered with fewer sensing features than common wearable and portable devices, there still exists hidden information in these processes reflecting the user's specific gait manner and motion statuses. Considering the particular movement pattern of the walking stick, we designed the linear-to-rotary structure (Figure 1c) to convert the low-frequency linear motion into a high-speed rotation. To track the hidden features of each striking, we further equipped the structure with three functional parts to form a hybridized unit, as illustrated in Figure 1b(i–iv). A P-TENG with five individual electrodes is assembled on top of the hybridized unit, to detect the contact point, sequence, and force of each strike. An EMG is introduced in the middle rotatory ratchet to harvest the wasted biomechanical energy for future self-sustainable systems. Below the EMG lies an R-TENG, which is utilized to detect gait abnormality from the irregular rotation motions. For self-powered TENG sensors, they are working based on the coupling effect of contact electrification[74] (triboelectrification) to convert the mechanical stimuli to electric output,[75] while the output signal is relative to the potential differences of applied materials,[76] contact force and speed,[77] contact area,[78] *etc.* And the components and working mechanisms for each part are discussed in detail in Note S1 (Supporting Information). Besides, to further increase the efficiency in harvesting biomechanical energy with ultralow frequency, a rotational unit with only an EMG is also developed (Figure 1b(v)). The rotational unit comprises a noncontacting ratchet design with rotation friction only coming from the bottom bearing, which further extends the rotation time to fill in the time gaps without external driving.

To achieve the linear-to-rotary transmission and fully utilize the bidirectional linear motion, one of the critical elements is the inner pawl. It is utilized to drive the outer ratchet to rotate and enable continuous rotating with inertia, depicted in Figure 1c(ii) (photos shown in Figure S3, Supporting Information). The pawl is rigidly linked to a circular disk through the top bearing's inner wall (Figure 1c(i)), which means the pawl will always have the same rotation status as the top circular disk. There are two claws in the pawl, which can be rotated freely because they are not rigidly fixed to the pawl frame. When the circular disk is rotated clockwise, the pawl will be rotated in the same way, and the two small claws with one side attached to it will be spread out due to centrifugal force (Figure 1c(ii)). The claws will be folded up when the circular disk is rotated anticlockwise due to the same reason. To convert the linear movement into ratchet rotation, a twist rod with patterns collaborative with the dumbbell-shaped hole at the circular disk center was designed (Figure S4, Supporting Information). One complete driving process contains four steps achieved by the pawl–ratchet system (Figure 1c(iii)). First, when an external pressure is applied to this device, the twist rod fixed on the lid will be moved downward and drive the circular disk to rotate clockwise. Claws will be spread out in this process and further embed into the ratchet's inner indentation to drive the outer ratchet to rotate, creating a linkage between the ratchet's rotation speed and the linear-driven speed. The twist rod will undergo an accelerating process moving downward, and it will reach the maximum speed when the contact to the bottom side is about to happen. After the twist rod reaches the bottom and stops moving, the freely linked claws will be folded up through the pushing of the ratchet's continuing rotation. Therefore, the ratchet is allowed to rotate for a certain time with its rotation inertia, free from the restriction of the pawl. Moreover, when the external stimulus is released, the twist rod will be moved upward driven by the tower-shaped spring placed between the casing and the lid. In this process, the pawls will be folded up more tightly due to the centrifugal force; thus the reverse linear motion will not cause a reverse driving force and hinder the rotation. It is noticed that in the last two steps stopping the external stimulus will not affect the continuous rotating of the ratchet, sparing the device from being limited by the stimuli frequencies as in previous designs,[79] hence contributing to a significant advancement in fully using human motion with ultralow frequency.

Ultra-Low-Frequency Biomechanical Energy Harvesting. With the linear-to-rotary structure, high-output biomechanical energy harvesting can be achieved through the designed hybridized unit. The output voltages of its the three main functional parts are first measured under three stimuli levels (Figure 2a–c), while their corresponding output current data are shown in Figure S5 (Supporting Information). In a motion process of the walking stick, it will first drive the ratchet to rotate with increasing speed in the contact process, resulting in an increased output of the EMG and R-TENG. When the walking stick reaches the maximum displacement and stops driving the ratchet, the rotation speed will reach the maximum value. Although the external driving is removed after that, the ratchet can still keep rotating for a certain time due to its rotation inertia. The output curves for three parts under medium stimuli are enlarged in the figure for clarity. Comparing output performances under three stimuli levels, it can be noticed that a higher stimuli will cause higher maximum output voltages for each part. Meanwhile, the ratchet will rotate for a longer time due to higher initial rotation speed. The maximum output voltage and peak output power for each part with different load resistances are shown in Figure 2d–f. For the P-TENG, EMG, and R-TENG, their maximum peak power reaches 2.98 mW, 6.22 mW, and 61.4 μW at 35 MΩ, 70 Ω, and 37.5 MΩ, respectively. The three parts are further connected to a rectifier bridge and linked with each other parallelly to charge an external load together (Figure 2g). The charging curve for a 10 μF capacitor is drawn in Figure 2h,

Downloaded from pubs.acs.org/ancac3/article-pdf/15/12/19054/1609523/nn1c04464.pdf by VELLORE INSTITUTE TECHNOLOGY user on 25 October 2026

https://doi.org/10.1021/acsnano.1c04464
ACS Nano 2021, 15, 19054–19069

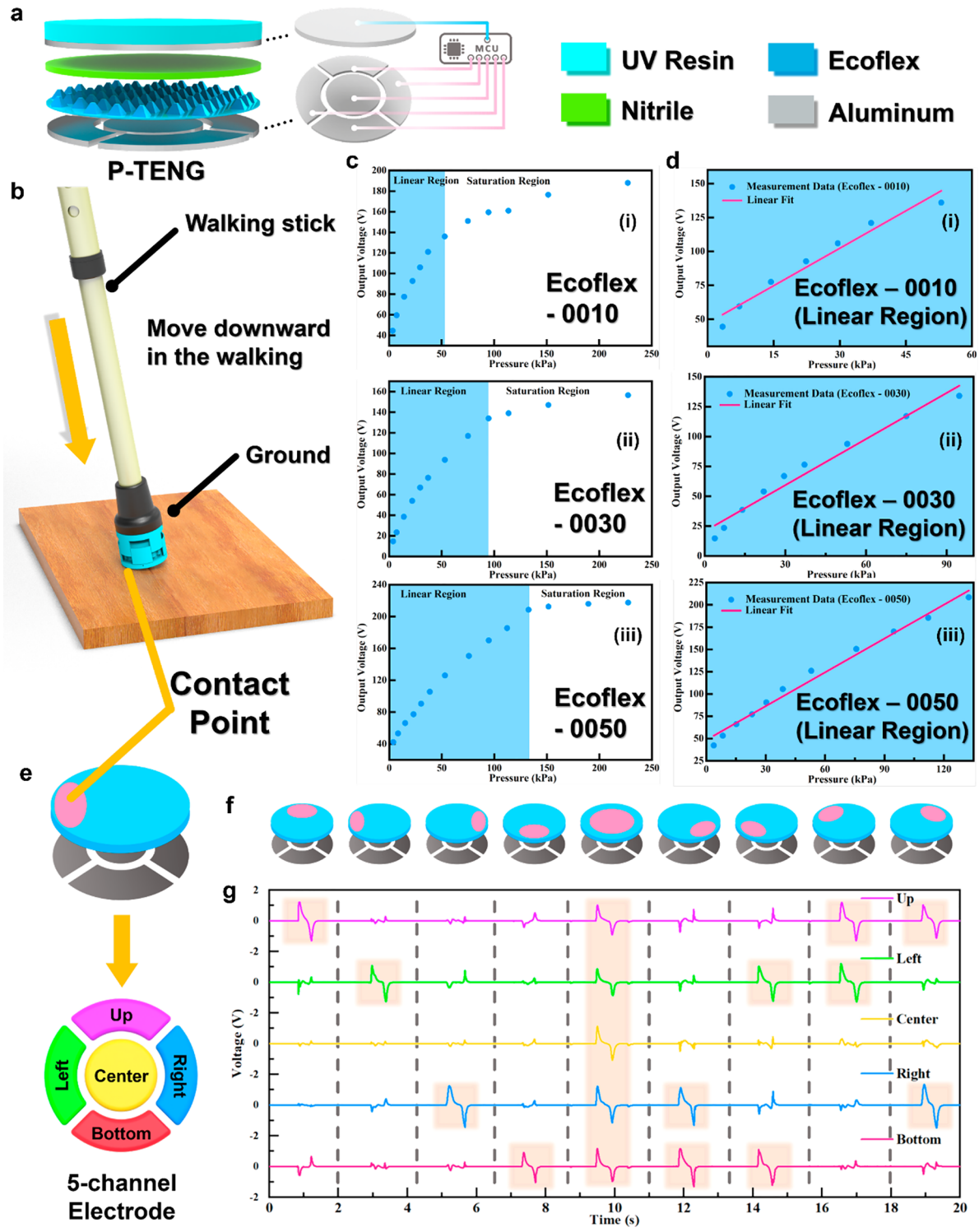


Figure 3. Characterization of the P-TENG for contact point detection. (a) Schematic of the electrical linkage for characterization of the P-TENG. (b) Schematic of a walking stick contacting the ground. (c) Characterizations of the output voltage under various applied forces for three different Ecoflex materials: (i) Ecoflex-0010, (ii) Ecoflex-0030, (iii) Ecoflex-0050. (d) Zoom-in of the data in the linear region and linear fitting curve for three different Ecoflex materials: (i) Ecoflex-0010, (ii) Ecoflex-0030, (iii) Ecoflex-0050. (e) Schematic drawing of the contact point and five bottom electrodes labeled with Up, Left, Center, Right, and Bottom, together with a specific color to represent each electrode. (f) Schematic drawing of nine typical contact points of the P-TENG with the ground. (g) Corresponding output curve of the five-channel P-TENG, with the corresponding output curve marked with a light pink background.

which shows the charging voltage can be significantly improved with the help of the R-TENG and P-TENG.

Although triboelectrification enables high-voltage output and essential self-powered sensing, it will also lead to a decrease of the rotation time caused by the friction. Besides, a single sensory unit would be enough for a walking stick to extract the essential information. Therefore, a rotational unit is developed to utilize the ultra-low-frequency kinetic energy

https://doi.org/10.1021/acsnano.1c04464
ACS Nano 2021, 15, 19054−19069

Downloaded from pubs.acs.org/ancac3/article-pdf/15/12/19054/1609523/nn1c04464.pdf by VELLORE INSTITUTE TECHNOLOGY user on 25 October 2026

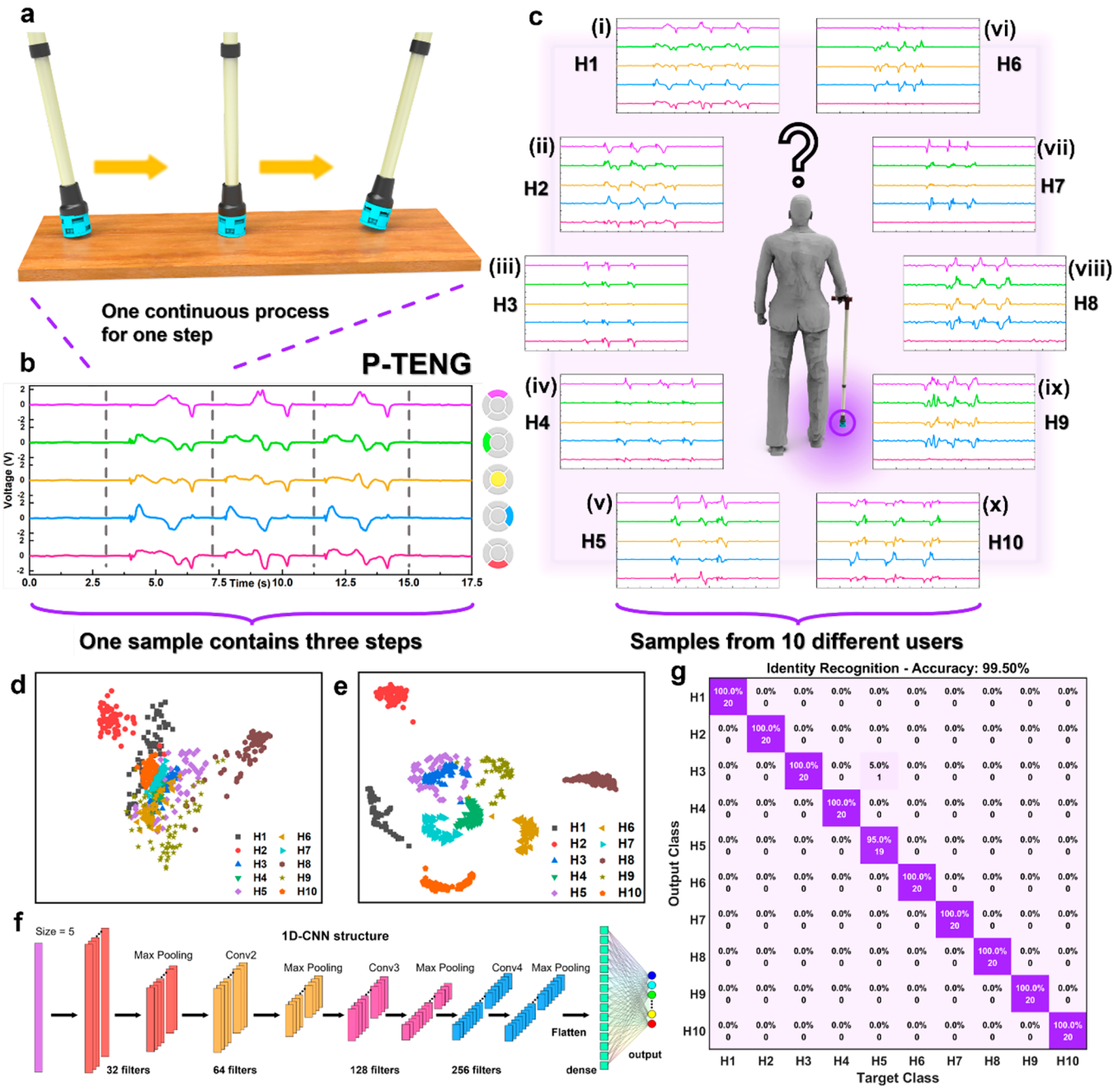


Figure 4. Identity recognition of the P-TENG enabled by deep learning. (a) Schematic of one contact and leaving process of the walking stick for one step. (b) Corresponding output curve generated from three steps of a user. (c) Output curves collected from 10 different users. All the curves have the same *x*-axis and *y*-axis as (b). The human model is reproduced with permission from turbosquid.com. (d) PCA from 10 different users. (e) t-SNE plot from the data set obtained from 10 different users. The separation of clusters corresponding to each user illustrates the discriminative capability of the P-TENG. (f) Detailed structure of the applied 1D-CNN model for deep learning. (g) Confusion matrix for the identity recognition for 10 different users.

more efficiently. It could be integrated with the hybridized unit by stacking them together in one walking stick, allowing for multimodal sensing and energy harvesting simultaneously (Figure 2i). According to previous research, the typical step frequency lies in the range of 84.7 to 136.4 steps/min for elderly people who can walk normally in the age range from 64 to 85 years old, while the frequency can be even lower for people with mobility disabilities.[80] Based on this, the output voltages of a rotational unit under various step frequencies ranging from 10 to 120 steps/min are measured (Figure S6, Supporting Information), and three of them are shown in Figure 2j. Compared with the output of the hybridized unit, a longer rotational time is obtained. The maximum output voltage and peak output power for this hybridized unit under various load resistances are shown in Figure 2k, which reaches a maximum peak power of 55.1 mW under a load resistance of 100 Ω. In practical walking, the device can be pushed one time every two steps, meaning the operational frequency is half of the step frequency. To demonstrate its advantages in that the rotational unit can still generate output continuously during the time intervals without external stimuli, the corresponding average energy (E) harvested in 1 min under various driving

https://doi.org/10.1021/acsnano.1c04464
ACS Nano 2021, 15, 19054−19069

Downloaded from pubs.acs.org/ancac3/article-pdf/15/12/19054/1609523/nn1c04464.pdf by VELLORE INSTITUTE TECHNOLOGY user on 25 October 2026

frequencies ranging from 10 steps/min [1/12 (0.083) Hz] to 120 steps/min (1 Hz) has been calculated through the following equation:

$$E = \int_t \frac{V^2}{R} dt \quad (1)$$

where V represents the output voltage, R represents the load resistance, and t is the time. The results are shown in Figure 2l. In 1 min, the harvested energy under 1 Hz is 1.65 J, which still remains 0.38 J when the driving frequency is decreased to only 0.083 Hz. Further dividing by the time period, the average output power can also be obtained:

$$P_{av} = \frac{E}{T} \quad (2)$$

where P_{av} represents the average output power in the time period T, which is 60 s, shown in Figure 2l. The average output power is 27.5 and 6.3 mW under 1 and 0.083 Hz, respectively. Considering the size of the rotational unit (⌀ 4.1 cm × 3.7 cm), the corresponding average power density driven under 1 and 0.083 Hz is 0.595 and 0.137 mW cm^{-3}, respectively. The comparison of the average power density between this rotational unit and other reported works aiming at scavenging biomechanical energy,[58,59,81−87] including flexible designs and nonflexible designs, is shown in Figure 2m. From this figure, it can be noticed that the rotational unit has the most significant average power density under the driving frequency of 1 Hz and can maintain a high average power density with ultralow frequencies. A more detailed and comprehensive comparison is shown in Table S1 (Supporting Information). On account of the specific characteristic of a walking stick, the potential for further enhancing the output performance through stacking units is illustrated through the charging curve for a 4 mF capacitor (Figure 2n). The stacking of two rotational units and one hybridized unit can charge this capacitor to 5 V in 8 s, under 1 Hz driving frequency, and the available space of the walking stick makes it feasible to stack more units to boost the output power further.

Deep-Learning-Enabled User Identification. To achieve intelligent real-time monitoring for elderly and motion-impaired people through a walking stick, a sensor with the ability to record the features of its motion is required. Compared to traditional MEMS inertial sensors, self-powered TENG sensors have the advantages of low power consumption and low data volume. With this in mind, a P-TENG with a five-electrode design is developed (Figure 3a). Figure 3b depicts the status of a walking stick that is about to contact the ground. The pressure distribution on the P-TENG will vary with the contact position, which will lead to a specific output of the five electrodes. To choose the best material suited for the application of the caregiving walking stick, we first tested three Ecoflex materials with different levels of softness, namely, the Ecoflex-0010, Ecofelx-0030, and Ecoflex-0050. Figure 3c(i−iii) show their open-circuit output voltages under different applied forces measured with the whole setup shown in Figure S7 (Supporting Information). The detailed output curves for some data points are shown in Figure S8 (Supporting Information). A larger force will lead to a larger contact area between the Ecoflex layer and the nitrile layer due to the deformation of the small frustums on the surface and lead to a higher output voltage. However, the value will saturate to a certain level depending on the softness of the applied materials (Figure 3d(i−iii)). Compared to the other two materials, the hardest, Ecoflex-0050, has the largest linear sensing range, which is able to sense applied pressure from 0 to 135 kPa. Considering the area of the P-TENG (⌀ 4.1 cm, 13.2 cm^2), the linear pressure sensing range for the above three materials means maximum sensing forces of 75, 125, and 175 N for Ecoflex-0010, 0030, and 0050, respectively. In the practical working conditions of the walking stick, the maximum applied force lies in the range of 10% to 20% body weight.[88] In other words, the walking stick needs to maximally bear forces ranging from 80 to 160 N for people with an 80 kg body weight. Based on the above medical data, the Ecoflex-0050 material was utilized to cover the force range of a walking stick in practical working conditions. We label the five bottom electrodes as Up, Left, Center, Right, and Bottom, respectively, as depicted in Figure 3e. Nine typical contact points of the device are drawn in Figure 3f, and their corresponding output voltages from each electrode are shown in Figure 3g. The contact point that undergoes the contact and separation process will lead to a large output of the electrodes beneath it, which is marked with a light pink background in Figure 3g. For example, when the walking stick contacts the ground with the area on top of the "Up" electrode, the "Up" electrode will have a corresponding significant output.

With the rapid development of AI technology, more devices and systems have been integrated with AI to realize intelligent decision-making, status recognition, and automation control. Through training an end-to-end neural network, devices become able to learn more representative features of raw signal data, thus enabling more advanced functionalities. Therefore, to achieve a caregiving walking stick with intelligent monitoring functions, we introduced the deep-learning method to analyze and extract all features of the five-channel P-TENG output. In a continuous process of a walking stick contacting and leaving the ground (Figure 4a), a specific output curve for the five-electrode P-TENG can be obtained (Figure 4b). In an entire gait cycle involving the walking stick, distinct output curves from different users and motions can be generated originating from the varying contact points, contact sequence, and contact force. Acquired by an Arduino MEGA 2560 microcontroller, the real-time output signals of the P-TENG for 10 different users are shown in Figure 4c(i−x), marked from H1 to H10. To build the whole data set, 80 samples for each user were collected, and one single sample contains output signals for three continuous steps with a total of 3500 data points for each channel. In these 80 samples, 60 sets of them are randomly picked for training and 20 sets for testing. To intuitively reveal the discriminative capacity of the P-TENG, high-dimensional sensory responses have been projected into 2D space *via* the principal component analysis (PCA) and t-distributed stochastic neighbor embedding (t-SNE), as shown in Figure 4d and e. It can be observed that the data collected from different users naturally form distinctive clusters, proving the walking stick as a reliable medium for identity distinguishing. While variations of the signals under different scenarios can be observed, it is hardly possible to extract enough features manually to distinguish these testing conditions. A deep-learning model was therefore built based on the one-dimensional convolutional neural network (1D-CNN) structure depicted in Figure 4f, to realize advanced data analytics. The corresponding confusion map for identity recognition of the 10 users is shown in Figure 4g, which shows a high accuracy of 99.5%. This high recognition accuracy has further verified that the gait patterns and motion

Downloaded from pubs.acs.org/ancac3/article-pdf/15/12/19054/1609523/nn1c04464.pdf by VELLORE INSTITUTE TECHNOLOGY user on 25 October 2026

https://doi.org/10.1021/acsnano.1c04464
ACS Nano 2021, 15, 19054−19069

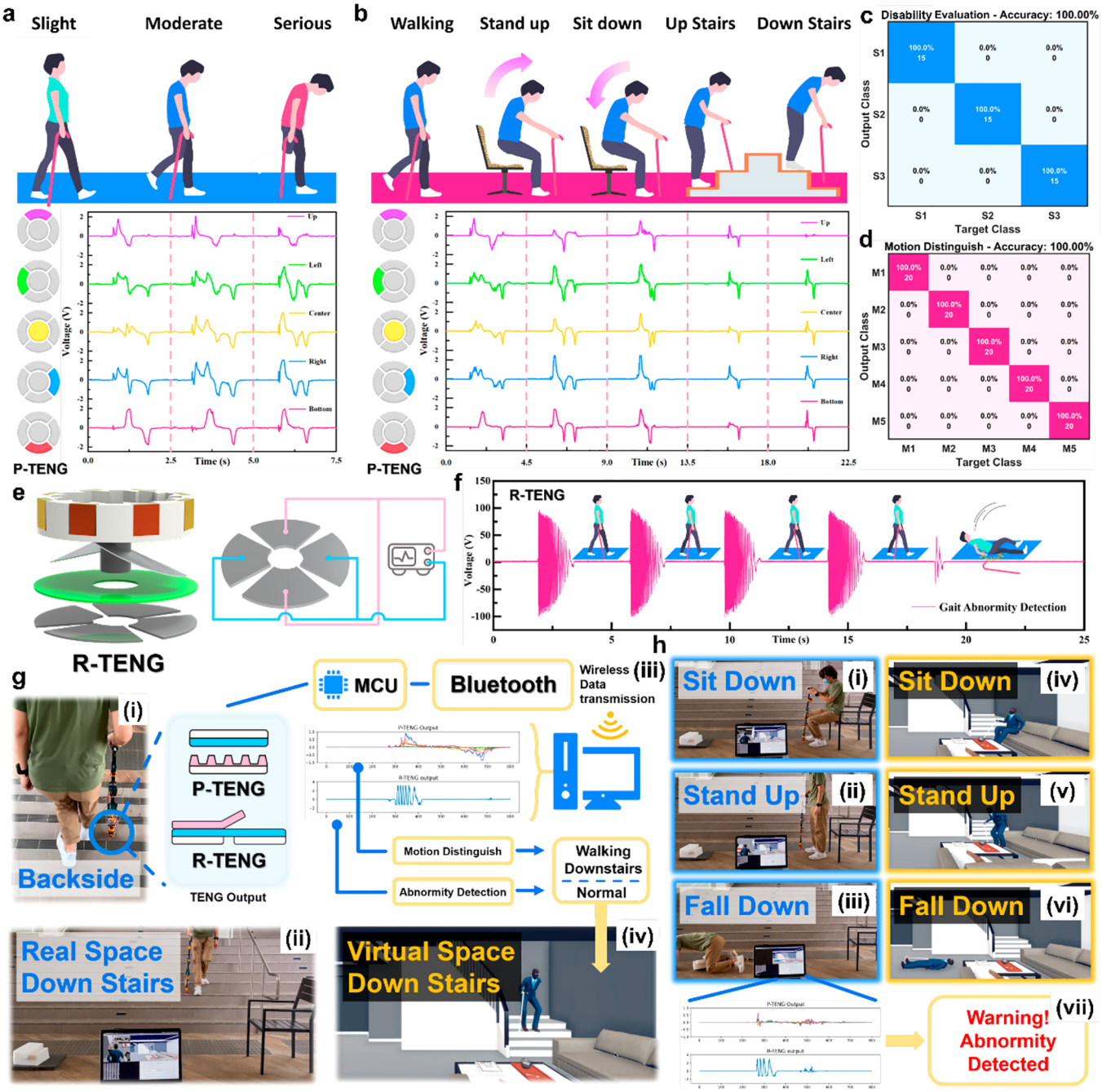


Figure 5. Real-time indoor monitoring functions of the caregiving walking stick enabled by deep learning. (a) Schematic of a user with three different levels of mobility disabilities and the corresponding output curve of the five-channel P-TENG. (b) Schematic of a user with moderate mobility disability under five different motion statuses and the corresponding output curve of the five-channel P-TENG. (c) Confusion matrix for the three-level abnormality evaluation. (d) Confusion matrix to distinguish the five motion statuses. (e) Schematic of electrode linkage of the R-TENG. (f) Output curve for the R-TENG under normal walking status and falling. (g) Demonstration for the application of indoor monitoring with the caregiving walking stick: (i) backside view showing a person is walking downstairs, (ii) photo taken from real space, (iii) whole setup showing the signals generated from the P-TENG and R-TENG will be first collected by an Arduino mega 2560, then be sent wirelessly to the computer through Bluetooth and be analyzed to obtain the real-time motion status of the user. (h) A person (i) sitting down, (ii) standing up, and (iii) falling down in real space with virtual elderly people (iv) sitting down, (v) standting up, and (vi) falling down in the virtual space, (vii) a warning will be sent out to summon help when detecting an abnormality. The human model is reproduced with permission from turbosquid.com.

characteristics of sensory information collected from the walking stick used by various users are distinct, although this phenomenon has rarely been investigated and discussed. Meanwhile, it also indicates that the five-electrode sensor design of the P-TENG is beneficial for subtle motion variation sensing, which is adequately sensitive enough to distinguish 10 movement patterns from different participants.

Deep-Learning-Based Multifunctional Indoor Monitoring. For elderly and motion-impaired people, monitoring their regular motions in an indoor environment such as home would be important to track their well-being statuses when

https://doi.org/10.1021/acsnano.1c04464
ACS Nano 2021, 15, 19054−19069

Downloaded from pubs.acs.org/ancac3/article-pdf/15/12/19054/1609523/nn1c04464.pdf by VELLORE INSTITUTE TECHNOLOGY user on 25 October 2026

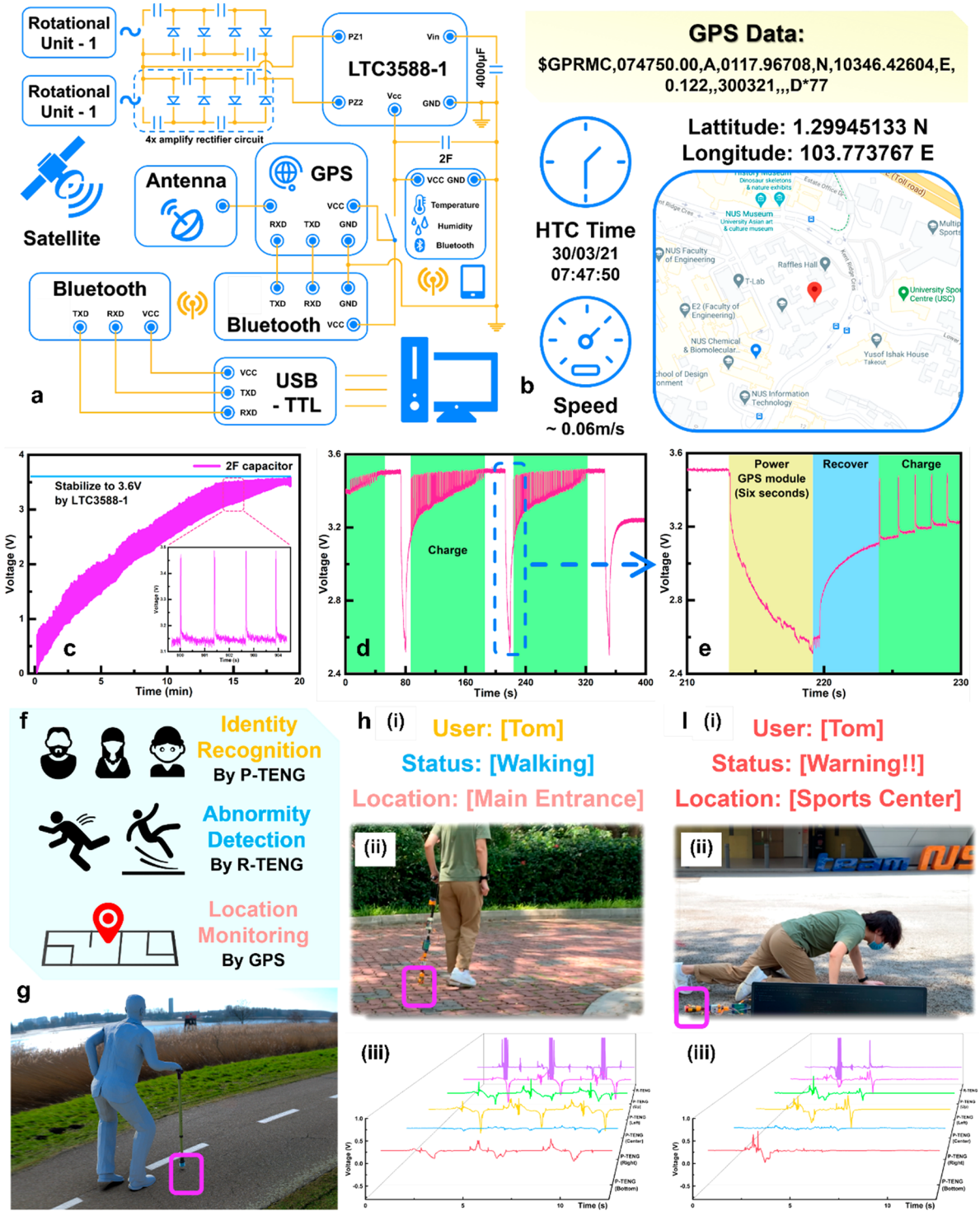


Figure 6. Self-sustained IoT sensing and demonstration for omnimonitoring applications. (a) Schematic of the proposed IoT system powered by two power units with a GPS location sensor, environmental temperature and humidity sensor, and Bluetooth wireless data transmission units. (b) Detailed information in one GPS signal sent per second, including time, location, and speed. (c) Charging curve of the 2 F supercapacitor with the LTC-3588 power management unit powered by two power units. (d) Charging and discharging curve of the 2 F supercapacitor after been charged and stabilized to around 3.6 V. (e) Zoom-in curve containing powering, self-recovering, and charging. (f) Schematic showing the main functions for outdoor applications of the caregiving walking stick. (g–i) Demonstration of a person using this caregiving walking stick walking in a park (h): (i) Monitoring system showing the real-time information obtained through the walking stick, including the user identification, motion status, and location, (ii) the user in real space with the caregiving walking stick, (iii) the output of the R-TENG and P-TENG when the user is normally walking. (i): (i) Monitoring system showing a gait abnormality has been detected in a detailed location and asking for immediate physical aid, (ii) the user falling in real space with the caregiving walking stick, (iii) the output of the R-TENG and P-TENG when the user has fallen down. The human model is reproduced with permission from turbosquid.com.

Downloaded from pubs.acs.org/ancac3/article-pdf/15/12/19054/16095223/nn1c04464.pdf by VELLORE INSTITUTE TECHNOLOGY user on 25 October 2026

https://doi.org/10.1021/acsnano.1c04464
ACS Nano 2021, 15, 19054−19069

they are staying alone. Meanwhile, their privacy should also be taken into consideration, which is a general concern for camera-based monitoring systems. The walking stick endowed with the essential sensing features can serve as a reliable tool for indoor caregiving of such users. Figure 5a shows the corresponding output curves of three persons with different mobility-impaired levels from the P-TENG, which are slightly impaired gait, moderate mobility disability, and serious mobility disability. Figure 5b shows the output curves of a person with a moderate mobility disability under five different motions, including walking, standing up, sitting down, walking upstairs, and walking downstairs. Although subtle differences can be observed between the five-channel outputs under different scenarios, it is still hard to categorize them based on the observable features. To fully reveal the potential of the P-TENG, deep-learning-based analytics is introduced here to extract hidden features for precise differentiation. In terms of the deep-learning data for mobility disability evaluation, the signal data from each channel is recorded with 3000 data points, containing three steps in one data sample. Sixty-five sets of data samples were collected, with a total of 15 000 × 65 data points, in which 50 sets of data samples were used for model training and 15 sets of data samples were used for testing. Three different levels of mobility disability, namely, slight, moderate, and serious, are labeled as S1, S2, and S3 in the confusion map shown in Figure 5c, with 100% accuracy. In terms of the deep-learning data for motion detection, one step is recorded for each sample, with 1500 data points for every channel. Eighty sets of data samples were collected; 60 of them were utilized for training and 20 sets were applied for testing, which also achieved 100% accuracy with a confusion map shown in Figure 5d. The high detection accuracies reveal additional functionalities of the walking stick apart from the aforementioned identity recognition, which would be helpful to achieve comprehensive monitoring of users such as the elderly and motion-impaired people.

At the same time, as another functional part in the proposed hybridized unit, the R-TENG can also serve as a significant monitoring window for users. The R-TENG is utilized to measure the strike speed of the walking stick in contact with the ground, whose irregular output pattern would imply an abnormal motion status. Its bottom electrode has been cut into four fan-shaped structures, and each two in the opposite direction are connected to form a single output (Figure 5e). The R-TENG shows great stability and accuracy in measuring linear speed, which is discussed in detail in Note S4 and Figure S11 (Supporting Information). The linear strike speed of the walking stick can reveal the health status of users to some extent, which could be further investigated in future work. Here, we mainly apply the R-TENG part to detect the gait abnormality of users, with schematics and corresponding output curves in Figure 5f. When the user is moving normally with this caregiving walking stick, typical outputs of the R-TENG can be obtained with a continuous high-speed rotation. However, when there is an abnormality such as a fall, irregular patterns of output with low intensity will be detected. In the practical working conditions, to avoid potential confusion with other similar movements such as accidentally dropping or consciously putting down or picking up the walking stick, the abnormality will be decided based on the joint output from the P-TENG and R-TENG (Figure S12, Supporting Information). Once this abnormal signal has been detected and no other normal signals have been generated after that, we can assume that the user has endured an accident and needs timely physical aid.

Combining the abnormality detection function achieved through the R-TENG and advanced recognition functions realized by the P-TENG, multifunctional real-time monitoring of users is applicable with this caregiving walking stick in various environments. We picked the most typical application scenario, indoor monitoring, to demonstrate the practical usage of the caregiving walking stick, as shown in Figure 5g. A virtual environment (Figure 5g(iv)) mimicking real life (Figure 5g(ii)) is built to reflect the real-time motion status of a person using the caregiving walking stick (Figure 5g(i)). The output voltage signals from the R-TENG and P-TENG are first collected by the Arduino mega 2560, then sent to the computer through a Bluetooth module for data processing and analysis, as shown in Figure 5g(iii). The exact motion can be detected through the trained deep-learning model mentioned above. Simultaneously, the stable output of the R-TENG also reflects a normal motion status. A video demonstrating the walking stick's real-time motion monitoring capability is provided in Video S1 (Supporting Information), where a subject imitates five representative motions of aged people using this caregiving walking stick, namely, walking downstairs, walking, sitting down, standing up, and falling down. Besides, the real-life images and corresponding virtual space are shown in Figure 5h. When the user falls down, an abnormal signal from the R-TENG will be received, implying an urgent need for help (Figure 5h(vii)). Unlike traditional camera-based indoor monitoring, which generally involves privacy concerns raised by video taking, this caregiving walking stick only monitors the user's motion status as an important well-being indicator. Moreover, the created virtual space also provides an intuitive interface to monitor the motion status of users without invading their privacy.

Omnimonitoring Empowered by a Self-Sustained IoT System. By virtue of the high power output of the rotational unit and the ability for layer stacking in the walking stick, a self-sustainable IoT system can be achieved through integrating power management units, energy storage units, and wireless data transmission units. To better assist motion-impaired users, especially in an outdoor environment, extrasensory information such as location and environmental amenity level would be desirable for a more capable caregiving walking stick. With this demand, we choose a commercial global positioning system (GPS) location-sensing module and an environmental temperature and humidity IoT module as the functional units for the self-sustained IoT system. The overall IoT system is depicted in Figure 6a. The GPS module can obtain one GPRMC data sentence per second from available satellites, which includes data on the current time, date, moving speed, and location, as shown in Figure 6b. First, to fully use the continuous but decreasing output caused by rotation friction, a pair of 4× amplify rectifier circuits were applied for each rotational unit and linked in parallel with each other. The output is further connected to a commercial power management unit (LTC-3588) for voltage stabilization. The output of the LTC-3588 is first collected by a 4 mF small capacitor. When the voltage of the small capacitor reaches 4 V, it automatically releases its energy to power a 2 F supercapacitor. Under a driving frequency of 1 Hz, it takes about 18 min to power this 2 F supercapacitor to near 3.6 V by two rotational units, as shown in Figure 6c. The apparent thickness of the charging curve is formed by the oscillations arising from the

Downloaded from pubs.acs.org/ancac3/article-pdf/15/12/19054/1609523/nn1c04464.pdf by VELLORE INSTITUTE TECHNOLOGY user on 25 October 2026

https://doi.org/10.1021/acsnano.1c04464
ACS Nano 2021, 15, 19054−19069

short-time high-voltage energy released from the small capacitor, which is applied to increase the charging efficiency for the supercapacitor. Meanwhile, in the charging process, this 2 F supercapacitor is continuously powering an IoT module with environmental temperature- and humidity-sensing functions. The corresponding received sensing data from this module are shown in Figure S13 (Supporting Information). After been charged and stabilized to near 3.6 V, the 2 F supercapacitor will be used to power the commercial GPS module (MAX-7Q) and a Bluetooth module, through which the GPRMC data can be sent wirelessly to a computer *via* a corresponding receiver for real-time location monitoring. The voltage on the supercapacitor in the cycling charging–discharging processes of the IoT system is shown in Figure 6d, in which the charging processes with two working rotational units are highlighted with a green background. Once the GPS module is connected to the supercapacitor, its voltage will rapidly decrease to 2.5 V in 6 s. This voltage is the minimum voltage required for powering the GPS module, which can send one complete GPRMC data sequence per second in this working mode. Although the voltage decreased drastically, it is mainly caused by the polarization of inner charges, while the energy stored in the supercapacitor actually still remains at a certain level. Therefore, after disconnecting the GPS module, the voltage can self-recover to near 3.2 V with the stabilization of stored charges. After that, the two rotational units working at 1 Hz now only need around 90 s to charge this supercapacitor back to 3.6 V and can power the whole IoT system for 6 s again. A detailed zoom-in figure depicting the draining, self-recovering, and charging curve is shown in Figure 6e. Overall, with the two rotational units working at 1 Hz, this IoT system can achieve continuously real-time environmental temperature and humidity sensing. At the same time, a self-sustained locating function is also realized, with a GPS module and a Bluetooth module working 6 s for every 90 s of charging.

By introducing the locating function, a comprehensive caregiving for elderly and motion-impaired people in outdoor environments becomes applicable. With the help of deep learning, the monitoring system can recognize the user's identity, real-time motion status, and corresponding location, as depicted in Figure 6f. In the coming 5 G era, the collected data can be sent to the cloud through the complete 5 G network for further analysis. It may be accessible to authorized users such as caregivers, who would be able to remotely track the well-being status and provide prompt assistance once accidents were detected. Figure 6g envisions an aged person using this caregiving walking stick walking in a park, though alone on the road, being safe and tracked continuously to avoid undetectable hazards that are of wide concern for this population. To demonstrate the potential of the caregiving walking stick toward such applications, a video showing the functionalities of the walking stick in various scenarios is provided in Video S2 (Supporting Information), with two scenarios shown in Figure 6h,i. In this video, two users first enter a park with the caregiving walking stick. After collecting the P-TENG's output signals for three consecutive steps, the system can identify them as [Evelyn] and [Tom], respectively, and starts to monitor their motion status with the R-TENG. The output of the R-TENG and P-TENG for user Tom walking in the park is shown in Figure 6h(iii). Besides, based on the GPS module, the system can also tell that they entered the park through the [Main Entrance]. During the whole time [Tom] strolls in the park, the system can keep monitoring his motion status and location. When [Tom] trips over a branch on the ground at the [Sports Center], as shown in Figure 6i(ii), the system can timely detect this abnormality and seek immediate help.

CONCLUSION

This report investigates an AI-enabled caregiving walking stick with intelligent monitoring functions and self-sustained IoT sensing powered by ultra-low-frequency human motion as a healthcare platform for elderly and motion-impaired people. To enable a sensory walking stick, a modularly designed hybridized unit is proposed with the combination of an EMG and self-powered triboelectric sensors. The hybridized unit has basic capabilities including contact point sensing and gait abnormality detection. Through the introduction of the deep-learning analysis approach, multiple advanced functionalities including identity recognition (accuracy: 99.5%), mobility disability evaluation (accuracy: 100%), and motion status determination (accuracy: 100%) have also been achieved. To fulfill various requirements of users and push forward the progress toward self-sustainable IoT systems, the rotational unit is developed aiming at harvesting ultra-low-frequency biomechanical energy more efficiently. One rotational unit achieves a maximum peak power of 55.1 mW at 1 Hz and a significant average power density of 0.137 mW cm^{-3} at 0.083 Hz. By stacking multiple units together in the walking stick, we can further boost the output performance. With the stacking of two rotational units and one hybridized unit, a 4 mF capacitor is charged to 5 V in only 8 s. Besides, with stacking two rotational units for powering, a self-sustainable IoT sensing system with a GPS location sensor, environment temperature and humidity amenity sensor, power management circuit, and wireless transmission module has been developed. With the aforementioned multifunctional capabilities, demonstrations to show the comprehensive monitoring for users have been created. In the indoor environment, the motions of users can be recognized instantaneously through the stick without privacy concerns. An alarm will be sent out immediately if a gait abnormality is detected to summon urgent physical aid for the user who stays alone. The additional GPS sensor enables the location tracking of users in various outdoor environments, allowing for an omnimonitoring of the users with autonomy and safety even without a caregiver around. This multidimensional information has improved the traditional walking stick from a passive physical aid to an intelligent well-being aid with broadened features in user identity recognition, well-being status monitoring, and location tracking. Furthermore, this caregiving walking stick would avoid the risks of undetectable accidents and spare the inevitability of in-person nursing for people with impaired mobility, enabling a prompt alarm without the restriction of the activity sphere. Looking forward, the proposed caregiving walking stick as an intriguing platform for mobility-impaired users would raise the awareness of academia on this unnoticed demand and possibly change the lives of these users.

METHODS

Fabrication of the P-TENG. Three different Ecoflex materials (Ecoflex 0010, Ecoflex 0030, and Ecoflex 00050) are obtained through mixing a 1:1 weight ratio of part A and part B for each material. The obtained Ecoflex gel is then poured into a mold with frustum-shaped holes fabricated through 3D printing. After drying in

Downloaded from pubs.acs.org/ancac3/article-pdf/15/12/19054/1609523/nn1c04464.pdf by VELLORE INSTITUTE TECHNOLOGY user on 25 October 2026

https://doi.org/10.1021/acsnano.1c04464
ACS Nano 2021, 15, 19054–19069

an oven at 50 °C for 1 h, the Ecoflex materials are attached to an adhesive aluminum layer.

Fabrication of the R-TENG. First, a circular aluminum layer is cut into four separate parts attached on top of the pedestal, with the two parts placed facing each other linked through a wire. A circular PTFE layer is then attached to the bottom electrode layers. Two aluminum sheets are cut into a fan shape and are mounted on the bottom of the ratchet. The curved shape of the aluminum fans ensures the contact to the area between it and the PTFE layer while limiting the contact force, which will cause high friction.

Fabrication of the EMG. Eight magnets (Nd-Fe-B-N52) are attached on the outer ratchet, with all of their south magnetic poles placed facing inside. Four copper coils with 500 turns are wrapped on the outside casing and connected in series with each other.

Characterization of the Output of the P-TENG. The voltage measurements for the P-TENG with a single-electrode design are carried out using an oscilloscope (Agilent, InfiniiVision, DSO-X 3034A) and electrometer (Keithley 6514) with a 10 MΩ probe, while the applied force is controlled by a force gauge. The generated triboelectric signals from the five-channel P-TENG are acquired by the signal acquisition module in an Arduino MEGA 2560 microcontroller in a real-time manner.

Characterization of the Output of the R-TENG. The voltage measurements for the R-TENG used for strike speed sensing and gait abnormality detection are carried out using an oscilloscope (Agilent, InfiniiVision, DSO-X 3034A) with a 100 MΩ probe.

Demonstration of the Caregiving Walking Stick. The signal acquisition module acquires the generated triboelectric signals of the R-TENG and P-TENG generated by different human motions through the Arduino MEGA 2560 microcontroller. The acquired electrical signals are sent to a laptop by a Bluetooth module (CC2530) instantly. Following that, the received signals are processed in Python with a trained CNN model also developed in Python with a Keras and TensorFlow backend. The recognized output is sent as a command to Unity 3D through TCP/IP to synchronize the movements in virtual space to that in real life.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acsnano.1c04464.

Video S1. Demonstration of the indoor healthcare monitoring with the caregiving walking stick (MP4)

Video S2. Demonstration of the omnimonitoring with the caregiving walking stick (MP4)

Notes S1−S5: designs and working principles of the hybridized unit; optimization strategy for the P-TENG; fabrication methods for each part; strike speed sensing by R-TENG; equations for calculating four parameters used for assessing the correlation between two parameters; Figures S1−13; and Table S1 (PDF)

AUTHOR INFORMATION

Corresponding Authors

Chengkuo Lee − *Department of Electrical & Computer Engineering, National University of Singapore, Singapore 117576, Singapore; National University of Singapore Suzhou Research Institute (NUSRI), Suzhou 215123, China; Center for Intelligent Sensors and MEMS (CISM), National University of Singapore, Singapore 117608, Singapore; NUS Graduate School−Integrative Sciences and Engineering Program (ISEP), National University of Singapore, Singapore 119077, Singapore;* orcid.org/0000-0002-8886-3649; Email: elelc@nus.edu.sg

Fei Wang − *School of Microelectronics, Southern University of Science and Technology, Shenzhen 518055, China;* orcid.org/0000-0003-1871-1022; Email: wangf@sustech.edu.cn

Authors

Xinge Guo − *Department of Electrical & Computer Engineering, National University of Singapore, Singapore 117576, Singapore; National University of Singapore Suzhou Research Institute (NUSRI), Suzhou 215123, China; Center for Intelligent Sensors and MEMS (CISM), National University of Singapore, Singapore 117608, Singapore; Institute of Microelectronics (IME), Agency for Science, Technology and Research (A*STAR), Singapore 138634, Singapore*

Tianyiyi He − *Department of Electrical & Computer Engineering, National University of Singapore, Singapore 117576, Singapore; National University of Singapore Suzhou Research Institute (NUSRI), Suzhou 215123, China; Center for Intelligent Sensors and MEMS (CISM), National University of Singapore, Singapore 117608, Singapore*

Zixuan Zhang − *Department of Electrical & Computer Engineering, National University of Singapore, Singapore 117576, Singapore; National University of Singapore Suzhou Research Institute (NUSRI), Suzhou 215123, China; Center for Intelligent Sensors and MEMS (CISM), National University of Singapore, Singapore 117608, Singapore*

Anxin Luo − *School of Microelectronics, Southern University of Science and Technology, Shenzhen 518055, China*

Eldwin J. Ng − *Institute of Microelectronics (IME), Agency for Science, Technology and Research (A*STAR), Singapore 138634, Singapore*

Yao Zhu − *Institute of Microelectronics (IME), Agency for Science, Technology and Research (A*STAR), Singapore 138634, Singapore*

Huicong Liu − *School of Mechanical and Electric Engineering, Jiangsu Provincial Key Laboratory of Advanced Robotics, Soochow University, Suzhou 215123, China*

Complete contact information is available at: https://pubs.acs.org/10.1021/acsnano.1c04464

Author Contributions

X.G. and C.L. conceived the idea. A.L. and F.W. helped to design and fabricate the device. X.G. designed and performed the experiments. X.G. took all the photos shown in the figures. X.G. and Z.Z. wrote the algorithms for machine learning and demonstration. X.G. and T.H. contributed to the data analysis and drafted the manuscript. X.G., C.L., E.N., Y.Z., and H.L. edited the manuscript.

Author Contributions

#X.G., T.H., Z.Z., and A.L. contributed equally to this work.

Funding

This work was supported by the research grant of National Key Research and Development Program of China (Grant No. 2019YFB2004800, Project No. R-2020-S-002) at NUSRI, Suzhou, China; the research grant of RIE Advanced Manufacturing and Engineering (AME) programmatic grant A18A4b0055 "Nanosystems at the Edge" at NUS, Singapore; Guangdong Natural Science Funds (2018A050506001); and Shenzhen Science and Technology Innovation Committee (JCYJ20200109105838951).

Notes

The authors declare no competing financial interest.

Downloaded from pubs.acs.org/ancac3/article-pdf/15/12/19054/1609523/nn1c04464.pdf by VELLORE INSTITUTE TECHNOLOGY user on 25 October 2026

REFERENCES

(1) Aging in Unity.. *Nat. Aging* **2021**, *1* (1), 1−1.
(2) World Bank Group. Disability Inclusion Overview. https://www.worldbank.org/en/topic/disability (accessed July 21, 2021).
(3) Chung, H. U.; Rwei, A. Y.; Hourlier-Fargette, A.; Xu, S.; Lee, K.; Dunne, E. C.; Xie, Z.; Liu, C.; Carlini, A.; Kim, D. H.; Ryu, D.; Kulikova, E.; Cao, J.; Odland, I. C.; Fields, K. B.; Hopkins, B.; Banks, A.; Ogle, C.; Grande, D.; Park, J. B.; et al. Skin-Interfaced Biosensors for Advanced Wireless Physiological Monitoring in Neonatal and Pediatric Intensive-Care Units. *Nat. Med.* **2020**, *26* (3), 418−429.
(4) Lee, G.-H.; Moon, H.; Kim, H.; Lee, G. H.; Kwon, W.; Yoo, S.; Myung, D.; Yun, S. H.; Bao, Z.; Hahn, S. K. Multifunctional Materials for Implantable and Wearable Photonic Healthcare Devices. *Nat. Rev. Mater.* **2020**, *5* (2), 149−165.
(5) Gao, W.; Emaminejad, S.; Nyein, H. Y. Y.; Challa, S.; Chen, K.; Peck, A.; Fahad, H. M.; Ota, H.; Shiraki, H.; Kiriya, D.; Lien, D.-H.; Brooks, G. A.; Davis, R. W.; Javey, A. Fully Integrated Wearable Sensor Arrays for Multiplexed *in Situ* Perspiration Analysis. *Nature* **2016**, *529* (7587), 509−514.
(6) Chu, B.; Burnett, W.; Chung, J. W.; Bao, Z. Bring on the BodyNET. *Nature* **2017**, *549* (7672), 328−330.
(7) Song, Y.; Min, J.; Yu, Y.; Wang, H.; Yang, Y.; Zhang, H.; Gao, W. Wireless Battery-Free Wearable Sweat Sensor Powered by Human Motion. *Sci. Adv.* **2020**, *6* (40), eaay9842.
(8) Gao, Y.; Yu, L.; Yeo, J. C.; Lim, C. T. Flexible Hybrid Sensors for Health Monitoring: Materials and Mechanisms to Render Wearability. *Adv. Mater.* **2020**, *32* (15), 1902133.
(9) Wu, F.; Qiu, C.; Wu, T.; Yuce, M. R. Edge-Based Hybrid System Implementation for Long-Range Safety and Healthcare IoT Applications. *IEEE Internet Things J.* **2021**, *4662* (c), 1−1.
(10) Wu, T.; Wu, F.; Qiu, C.; Redoute, J.-M.; Yuce, M. R. A Rigid-Flex Wearable Health Monitoring Sensor Patch for IoT-Connected Healthcare Applications. *IEEE Internet Things J.* **2020**, *7* (8), 6932−6945.
(11) Arab Hassani, F.; Shi, Q.; Wen, F.; He, T.; Haroun, A.; Yang, Y.; Feng, Y.; Lee, C. Smart Materials for Smart Healthcare− Moving from Sensors and Actuators to Self-Sustained Nanoenergy Nanosystems. *Smart Mater. Med.* **2020**, *1* (July), 92−124.
(12) Cho, Y.; Park, J.; Lee, C.; Lee, S. Recent Progress on Peripheral Neural Interface Technology towards Bioelectronic Medicine. *Bioelectron. Med.* **2020**, *6* (1), 23.
(13) Wang, J.; Wang, H.; Thakor, N. V.; Lee, C. Self-Powered Direct Muscle Stimulation Using a Triboelectric Nanogenerator (TENG) Integrated with a Flexible Multiple-Channel Intramuscular Electrode. *ACS Nano* **2019**, *13* (3), 3589−3599.
(14) Wang, J.; Wang, H.; He, T.; He, B.; Thakor, N. V.; Lee, C. Investigation of Low-Current Direct Stimulation for Rehabilitation Treatment Related to Muscle Function Loss Using Self-Powered TENG System. *Adv. Sci.* **2019**, *6* (14), 1900149.
(15) Wang, J.; He, T.; Lee, C. Development of Neural Interfaces and Energy Harvesters towards Self-Powered Implantable Systems for Healthcare Monitoring and Rehabilitation Purposes. *Nano Energy* **2019**, *65* (July), 104039.
(16) Shi, J.; Liu, S.; Zhang, L.; Yang, B.; Shu, L.; Yang, Y.; Ren, M.; Wang, Y.; Chen, J.; Chen, W.; Chai, Y.; Tao, X. Smart Textile-Integrated Microelectronic Systems for Wearable Applications. *Adv. Mater.* **2020**, *32* (5), 1901958.
(17) Dietz, M. A.; Goetz, C. G.; Stebbins, G. T. Evaluation of a Modified Inverted Walking Stick as a Treatment for Parkinsonian Freezing Episodes. *Mov. Disord.* **1990**, *5* (3), 243−247.
(18) Kessentini, O.; Dalce, R.; Megdiche, I.; Bastide, R. *Towards Predicting Frailty Symptoms through a Smart Walking Stick*. IFIP/IEEE International Conference on Performance Evaluation and Modeling in Wired and Wireless Networks (PEMWN), France, September 26−28; IEEE, 2018; pp 1−7; DOI: 10.23919/PEMWN.2018.8548901.
(19) Dabir, A.; Solkar, R.; Kumbhar, M.; Narayanan, G. *GPS and IoT Equipped Smart Walking Stick*. International Conference on Communication and Signal Processing (ICCSP), China, March 16−19; IEEE, 2018; pp 0322−0326; DOI: 10.1109/ICCSP.2018.8524472.
(20) Yuan, F.; Liu, S.; Zhou, J.; Wang, S.; Wang, Y.; Xuan, S.; Gong, X. Smart Touchless Triboelectric Nanogenerator towards Safeguard and 3D Morphological Awareness. *Nano Energy* **2021**, *86*, 106071.
(21) Chang, W. J.; Chen, L. B.; Chen, M. C.; Su, J. P.; Sie, C. Y.; Yang, C. H. Design and Implementation of an Intelligent Assistive System for Visually Impaired People for Aerial Obstacle Avoidance and Fall Detection. *IEEE Sens. J.* **2020**, *20* (17), 10199−10210.
(22) Sahoo, N.; Lin, H. W.; Chang, Y. H. Design and Implementation of a Walking Stick Aid for Visually Challenged People. *Sensors* **2019**, *19* (1), 130.
(23) Lee, K.; Ni, X.; Lee, J. Y.; Arafa, H.; Pe, D. J.; Xu, S.; Avila, R.; Irie, M.; Lee, J. H.; Easterlin, R. L.; Kim, D. H.; Chung, H. U.; Olabisi, O. O.; Getaneh, S.; Chung, E.; Hill, M.; Bell, J.; Jang, H.; Liu, C.; Park, J. B.; et al. Mechano-Acoustic Sensing of Physiological Processes and Body Motions *via* a Soft Wireless Device Placed at the Suprasternal Notch. *Nat. Biomed. Eng.* **2020**, *4* (2), 148−158.
(24) Liu, S. Q.; Zhang, J. C.; Zhu, R. A Wearable Human Motion Tracking Device Using Micro Flow Sensor Incorporating a Micro Accelerometer. *IEEE Trans. Biomed. Eng.* **2020**, *67* (4), 940−948.
(25) Liu, S.; Zhang, J.; Zhang, Y.; Zhu, R. A Wearable Motion Capture Device Able to Detect Dynamic Motion of Human Limbs. *Nat. Commun.* **2020**, *11* (1), 5615.
(26) Jeong, H.; Rogers, J. A.; Xu, S. Continuous On-Body Sensing for the COVID-19 Pandemic: Gaps and Opportunities. *Sci. Adv.* **2020**, *6* (36), eabd4794.
(27) Wen, F.; He, T.; Liu, H.; Chen, H.-Y.; Zhang, T.; Lee, C. Advances in Chemical Sensing Technology for Enabling the Next-Generation Self-Sustainable Integrated Wearable System in the IoT Era. *Nano Energy* **2020**, *78* (July), 105155.
(28) Zhou, H.; Zhang, Y.; Qiu, Y.; Wu, H.; Qin, W.; Liao, Y.; Yu, Q.; Cheng, H. Stretchable Piezoelectric Energy Harvesters and Self-Powered Sensors for Wearable and Implantable Devices. *Biosens. Bioelectron.* **2020**, *168* (May), 112569.
(29) Chu, Y.; Zhong, J.; Liu, H.; Ma, Y.; Liu, N.; Song, Y.; Liang, J.; Shao, Z.; Sun, Y.; Dong, Y.; Wang, X.; Lin, L. Human Pulse Diagnosis for Medical Assessments Using a Wearable Piezoelectret Sensing System. *Adv. Funct. Mater.* **2018**, *28* (40), 1803413.
(30) Dong, K.; Peng, X.; Wang, Z. L. Fiber/Fabric-Based Piezoelectric and Triboelectric Nanogenerators for Flexible/Stretchable and Wearable Electronics and Artificial Intelligence. *Adv. Mater.* **2020**, *32* (5), 1902549.
(31) Li, Z.; Zheng, Q.; Wang, Z. L.; Li, Z. Nanogenerator-Based Self-Powered Sensors for Wearable and Implantable Electronics. *Research* **2020**, *2020*, 1−25.
(32) Zhu, M.; Sun, Z.; Chen, T.; Lee, C. Low Cost Exoskeleton Manipulator Using Bidirectional Triboelectric Sensors Enhanced Multiple Degree of Freedom Sensory System. *Nat. Commun.* **2021**, *12* (1), 2692.
(33) Zhu, M.; Shi, Q.; He, T.; Yi, Z.; Ma, Y.; Yang, B.; Chen, T.; Lee, C. Self-Powered and Self-Functional Cotton Sock Using Piezoelectric and Triboelectric Hybrid Mechanism for Healthcare and Sports Monitoring. *ACS Nano* **2019**, *13*, 1.
(34) Zhu, J.; Sun, Z.; Xu, J.; Walczak, R. D.; Dziuban, J. A.; Lee, C. Volatile Organic Compounds Sensing Based on Bennet Doubler-Inspired Triboelectric Nanogenerator and Machine Learning-Assisted Ion Mobility Analysis. *Sci. Bull.* **2021**, *66* (12), 1176−1185.
(35) Zhu, J.; Zhu, M.; Shi, Q.; Wen, F.; Liu, L.; Dong, B.; Haroun, A.; Yang, Y.; Vachon, P.; Guo, X.; He, T.; Lee, C. Progress in TENG Technology—A Journey from Energy Harvesting to Nanoenergy and Nanosystem. *EcoMat* **2020**, *2* (4), 1−53.
(36) Gao, Q.; Cheng, T.; Wang, Z. L. Triboelectric Mechanical Sensors—Progress and Prospects. *Extrem. Mech. Lett.* **2021**, *42*, 101100.
(37) Kwak, S. S.; Yoon, H.-J.; Kim, S.-W. Textile-Based Triboelectric Nanogenerators for Self-Powered Wearable Electronics. *Adv. Funct. Mater.* **2019**, *29* (2), 1804533.

Downloaded from pubs.acs.org/ancac3/article-pdf/15/12/19054/1609523/nn1c04464.pdf by VELLORE INSTITUTE TECHNOLOGY user on 25 October 2026

https://doi.org/10.1021/acsnano.1c04464
ACS Nano 2021, 15, 19054−19069

(38) Liu, L.; Guo, X.; Lee, C. Nano Energy Promoting Smart Cities into the 5G Era with Multi-Field Internet of Things (IoT) Applications Powered with Advanced Mechanical Energy Harvesters. *Nano Energy* **2021**, *88* (July), 106304.
(39) Dong, B.; Shi, Q.; Yang, Y.; Wen, F.; Zhang, Z.; Lee, C. Technology Evolution from Self-Powered Sensors to AIoT Enabled Smart Homes. *Nano Energy* **2021**, *79*, 105414.
(40) Zhou, Y.; Shen, M.; Cui, X.; Shao, Y.; Li, L.; Zhang, Y. Triboelectric Nanogenerator Based Self-Powered Sensor for Artificial Intelligence. *Nano Energy* **2021**, *84*, 105887.
(41) Sun, Z.; Zhu, M.; Zhang, Z.; Chen, Z.; Shi, Q.; Shan, X.; Yeow, R. C. H.; Lee, C. Artificial Intelligence of Things (AIoT) Enabled Virtual Shop Applications Using Self-Powered Sensor Enhanced Soft Robotic Manipulator. *Adv. Sci.* **2021**, *2100230*, 2100230.
(42) Karan, S. K.; Maiti, S.; Lee, J. H.; Mishra, Y. K.; Khatua, B. B.; Kim, J. K. Recent Advances in Self-Powered Tribo-/Piezoelectric Energy Harvesters: All-In-One Package for Future Smart Technologies. *Adv. Funct. Mater.* **2020**, *30* (48), 2004446.
(43) Guo, X.; Liu, L.; Zhang, Z.; Gao, S.; He, T.; Shi, Q.; Lee, C. Technology Evolution from Micro-Scale Energy Harvesters to Nanogenerators. *J. Micromech. Microeng.* **2021**, accepted for publication. DOI: 10.1088/1361-6439/ac168e
(44) Bai, Y.; Jantunen, H.; Juuti, J. Energy Harvesting Research: The Road from Single Source to Multisource. *Adv. Mater.* **2018**, *30* (34), 1707271.
(45) Wang, Y.; Wang, H.; Xuan, J.; Leung, D. Y. C. Powering Future Body Sensor Network Systems: A Review of Power Sources. *Biosens. Bioelectron.* **2020**, *166* (July), 112410.
(46) Hudak, N. S.; Amatucci, G. G. Small-Scale Energy Harvesting through Thermoelectric, Vibration, and Radiofrequency Power Conversion. *J. Appl. Phys.* **2008**, *103* (10), 101301.
(47) Beeby, S. P.; Tudor, M. J.; White, N. M. Energy Harvesting Vibration Sources for Microsystems Applications. *Meas. Sci. Technol.* **2006**, *17* (12), R175−R195.
(48) Wen, Z.; Yeh, M.-H.; Guo, H.; Wang, J.; Zi, Y.; Xu, W.; Deng, J.; Zhu, L.; Wang, X.; Hu, C.; Zhu, L.; Sun, X.; Wang, Z. L. Self-Powered Textile for Wearable Electronics by Hybridizing Fiber-Shaped Nanogenerators, Solar Cells, and Supercapacitors. *Sci. Adv.* **2016**, *2* (10), e1600097.
(49) Han, K.; Kim, J.; Rajabi-Abhari, A.; Bui, V.; Kim, J.; Choi, D.; Oh, I. Long-Lasting and Steady Triboelectric Energy Harvesting from Low-Frequency Irregular Motions Using Escapement Mechanism. *Adv. Energy Mater.* **2021**, *11* (4), 2002929.
(50) Wang, J.; Li, S.; Yi, F.; Zi, Y.; Lin, J.; Wang, X.; Xu, Y.; Wang, Z. L. Sustainably Powering Wearable Electronics Solely by Biomechanical Energy. *Nat. Commun.* **2016**, *7* (1), 12744.
(51) Chen, G.; Li, Y.; Bick, M.; Chen, J. Smart Textiles for Electricity Generation. *Chem. Rev.* **2020**, *120* (8), 3668−3720.
(52) Guo, H.; Yeh, M.-H.; Lai, Y.-C.; Zi, Y.; Wu, C.; Wen, Z.; Hu, C.; Wang, Z. L. All-in-One Shape-Adaptive Self-Charging Power Package for Wearable Electronics. *ACS Nano* **2016**, *10* (11), 10580−10588.
(53) Huang, L.; Lin, S.; Xu, Z.; Zhou, H.; Duan, J.; Hu, B.; Zhou, J. Fiber-Based Energy Conversion Devices for Human-Body Energy Harvesting. *Adv. Mater.* **2020**, *32* (5), 1902034.
(54) Pu, X.; Hu, W.; Wang, Z. L. Toward Wearable Self-Charging Power Systems: The Integration of Energy-Harvesting and Storage Devices. *Small* **2018**, *14* (1), 1702817.
(55) Seung, W.; Gupta, M. K.; Lee, K. Y.; Shin, K.; Lee, J.; Kim, T. Y.; Kim, S.; Lin, J.; Kim, J. H.; Kim, S. Nanopatterned Textile-Based Wearable Triboelectric Nanogenerator. *ACS Nano* **2015**, *9* (4), 3501−3509.
(56) Hou, C.; Chen, T.; Li, Y.; Huang, M.; Shi, Q.; Liu, H.; Sun, L.; Lee, C. A Rotational Pendulum Based Electromagnetic/Triboelectric Hybrid-Generator for Ultra-Low-Frequency Vibrations Aiming at Human Motion and Blue Energy Applications. *Nano Energy* **2019**, *63* (March), 103871.
(57) Liu, L.; Shi, Q.; Lee, C. A Hybridized Electromagnetic-Triboelectric Nanogenerator Designed for Scavenging Biomechanical Energy in Human Balance Control. *Nano Res.* **2021**, *12* (1), DOI: 10.1007/s12274-021-3540-7.
(58) Maharjan, P.; Bhatta, T.; Cho, H.; Hui, X.; Park, C.; Yoon, S.; Salauddin, M.; Rahman, M. T.; Rana, S. S.; Park, J. Y. A Fully Functional Universal Self-Chargeable Power Module for Portable/Wearable Electronics and Self-Powered IoT Applications. *Adv. Energy Mater.* **2020**, *10* (48), 2002782.
(59) Maharjan, P.; Bhatta, T.; Salauddin Rasel, M.; Salauddin, M.; Toyabur Rahman, M.; Park, J. Y. High-Performance Cycloid Inspired Wearable Electromagnetic Energy Harvester for Scavenging Human Motion Energy. *Appl. Energy* **2019**, *256*, 113987.
(60) Quan, T.; Wang, X.; Wang, Z. L.; Yang, Y. Hybridized Electromagnetic−Triboelectric Nanogenerator for a Self-Powered Electronic Watch. *ACS Nano* **2015**, *9* (12), 12301−12310.
(61) Tan, P.; Zheng, Q.; Zou, Y.; Shi, B.; Jiang, D.; Qu, X.; Ouyang, H.; Zhao, C.; Cao, Y.; Fan, Y.; Wang, Z. L.; Li, Z. A Battery-Like Self-Charge Universal Module for Motional Energy Harvest. *Adv. Energy Mater.* **2019**, *9* (36), 1901875.
(62) Mitcheson, P. D.; Yeatman, E. M.; Rao, G. K.; Holmes, A. S.; Green, T. C. Energy Harvesting from Human and Machine Motion for Wireless Electronic Devices. *Proc. IEEE* **2008**, *96* (9), 1457−1486.
(63) Liu, H.; Tay, C. J.; Quan, C.; Kobayashi, T.; Lee, C. A Scrape-Through Piezoelectric MEMS Energy Harvester with Frequency Broadband and Up-Conversion Behaviors. *Microsyst. Technol.* **2011**, *17* (12), 1747−1754.
(64) Kulah, H.; Najafi, K. Energy Scavenging From Low-Frequency Vibrations by Using Frequency Up-Conversion for Wireless Sensor Applications. *IEEE Sens. J.* **2008**, *8* (3), 261−268.
(65) Yan, C.; Gao, Y.; Zhao, S.; Zhang, S.; Zhou, Y.; Deng, W.; Li, Z.; Jiang, G.; Jin, L.; Tian, G.; Yang, T.; Chu, X.; Xiong, D.; Wang, Z.; Li, Y.; Yang, W.; Chen, J. A Linear-to-Rotary Hybrid Nanogenerator for High-Performance Wearable Biomechanical Energy Harvesting. *Nano Energy* **2020**, *67*, 104235.
(66) Jiang, D.; Ouyang, H.; Shi, B.; Zou, Y.; Tan, P.; Qu, X.; Chao, S.; Xi, Y.; Zhao, C.; Fan, Y.; Li, Z. A Wearable Noncontact Free-Rotating Hybrid Nanogenerator for Self-Powered Electronics. *InfoMat* **2020**, *2* (6), 1191−1200.
(67) Luo, A.; Zhang, Y.; Dai, X.; Wang, Y.; Xu, W.; Lu, Y.; Wang, M.; Fan, K.; Wang, F. An Inertial Rotary Energy Harvester for Vibrations at Ultra-Low Frequency with High Energy Conversion Efficiency. *Appl. Energy* **2020**, *279*, 115762.
(68) Lin, Y.; Chen, J.; Tavakoli, M. M.; Gao, Y.; Zhu, Y.; Zhang, D.; Kam, M.; He, Z.; Fan, Z. Printable Fabrication of a Fully Integrated and Self-Powered Sensor System on Plastic Substrates. *Adv. Mater.* **2019**, *31* (5), 1804285.
(69) Kim, C. S.; Yang, H. M.; Lee, J.; Lee, G. S.; Choi, H.; Kim, Y. J.; Lim, S. H.; Cho, S. H.; Cho, B. J. Self-Powered Wearable Electrocardiography Using a Wearable Thermoelectric Power Generator. *ACS Energy Lett.* **2018**, *3* (3), 501−507.
(70) He, T.; Wang, H.; Wang, J.; Tian, X.; Wen, F.; Shi, Q.; Ho, J. S.; Lee, C. Self-Sustainable Wearable Textile Nano-Energy Nano-System (NENS) for Next-Generation Healthcare Applications. *Adv. Sci.* **2019**, *6* (24), 1901437.
(71) Zhang, Z.; He, T.; Zhu, M.; Sun, Z.; Shi, Q.; Zhu, J.; Dong, B.; Yuce, M. R.; Lee, C. Deep Learning-Enabled Triboelectric Smart Socks for IoT-Based Gait Analysis and VR Applications. *npj Flex. Electron.* **2020**, *4* (1), 29.
(72) Zhou, Z.; Chen, K.; Li, X.; Zhang, S.; Wu, Y.; Zhou, Y.; Meng, K.; Sun, C.; He, Q.; Fan, W.; Fan, E.; Lin, Z.; Tan, X.; Deng, W.; Yang, J.; Chen, J. Sign-to-Speech Translation Using Machine-Learning-Assisted Stretchable Sensor Arrays. *Nat. Electron.* **2020**, *3* (9), 571−578.
(73) Gao, S.; He, T.; Zhang, Z.; Ao, H.; Jiang, H.; L, C. A Motion Capturing and Energy Harvesting Hybridized Lower-Limb System for Rehabilitation and Sports Applications. *Adv. Sci.* **2021**.
(74) Kaponig, M.; Mölleken, A.; Nienhaus, H.; Möller, R. Dynamics of Contact Electrification. *Sci. Adv.* **2021**, *7* (22), eabg7595.
(75) Wang, Z. L.; Wang, A. C. On the Origin of Contact-Electrification. *Mater. Today* **2019**, *30*, 34−51.

Downloaded from pubs.acs.org/ancac3/article-pdf/15/12/19054/1609523/nn1c04464.pdf by VELLORE INSTITUTE TECHNOLOGY user on 25 October 2026

(76) Zou, H.; Zhang, Y.; Guo, L.; Wang, P.; He, X.; Dai, G.; Zheng, H.; Chen, C.; Wang, A. C.; Xu, C.; Wang, Z. L. Quantifying the Triboelectric Series. *Nat. Commun.* **2019**, *10* (1), 1427.
(77) Niu, S.; Wang, S.; Lin, L.; Liu, Y.; Zhou, Y. S.; Hu, Y.; Wang, Z. L. Theoretical Study of Contact-Mode Triboelectric Nanogenerators as an Effective Power Source. *Energy Environ. Sci.* **2013**, *6* (12), 3576−3583.
(78) Zi, Y.; Niu, S.; Wang, J.; Wen, Z.; Tang, W.; Wang, Z. L. Standards and Figure-of-Merits for Quantifying the Performance of Triboelectric Nanogenerators. *Nat. Commun.* **2015**, *6*, DOI: 10.1038/ncomms9376.
(79) Fu, H.; Mei, X.; Yurchenko, D.; Zhou, S.; Theodossiades, S.; Nakano, K.; Yeatman, E. M. Rotational Energy Harvesting for Self-Powered Sensing. *Joule* **2021**, 1−45.
(80) Zijlstra, A.; de Bruin, E. D.; Bruins, N.; Zijlstra, W. The Step Length - Frequency Relationship in Physically Active Community-Dwelling Older Women. *Eur. J. Appl. Physiol.* **2008**, *104* (3), 427−434.
(81) Yang, Y.; Sun, N.; Wen, Z.; Cheng, P.; Zheng, H.; Shao, H.; Xia, Y.; Chen, C.; Lan, H.; Xie, X.; Zhou, C.; Zhong, J.; Sun, X.; Lee, S.-T. Liquid-Metal-Based Super-Stretchable and Structure-Designable Triboelectric Nanogenerator for Wearable Electronics. *ACS Nano* **2018**, *12* (2), 2027−2034.
(82) Wang, L.; Liu, W.; Yan, Z.; Wang, F.; Wang, X. Stretchable and Shape-Adaptable Triboelectric Nanogenerator Based on Biocompatible Liquid Electrolyte for Biomechanical Energy Harvesting and Wearable Human−Machine Interaction. *Adv. Funct. Mater.* **2021**, *31* (7), 2007221.
(83) Fan, K.; Cai, M.; Wang, F.; Tang, L.; Liang, J.; Wu, Y.; Qu, H.; Tan, Q. A String-Suspended and Driven Rotor for Efficient Ultra-Low Frequency Mechanical Energy Harvesting. *Energy Convers. Manage.* **2019**, *198* (July), 111820.
(84) Wu, Z.; Tang, J.; Zhang, X.; Yu, Z. An Energy Harvesting Bracelet. *Appl. Phys. Lett.* **2017**, *111* (1), 013903.
(85) Saha, C. R.; O'Donnell, T.; Wang, N.; McCloskey, P. Electromagnetic Generator for Harvesting Energy from Human Motion. *Sens. Actuators, A* **2008**, *147* (1), 248−253.
(86) Halim, M. A.; Rantz, R.; Zhang, Q.; Gu, L.; Yang, K.; Roundy, S. An Electromagnetic Rotational Energy Harvester Using Sprung Eccentric Rotor, Driven by Pseudo-Walking Motion. *Appl. Energy* **2018**, *217*, 66−74.
(87) Galchev, T.; Aktakka, E. E.; Najafi, K. A Piezoelectric Parametric Frequency Increased Generator for Harvesting Low-Frequency Vibrations. *J. Microelectromech. Syst.* **2012**, *21* (6), 1311−1320.
(88) Hart, J.; Hall, M.; Wrigley, T. V.; Marshall, C. J.; Bennell, K. L. Body Weight Support through a Walking Cane in Inexperienced Users with Knee Osteoarthritis. *Gait Posture* **2019**, *67*, 50−56.

Downloaded from pubs.acs.org/ancac3/article-pdf/15/12/19054/1609523/nn1c04464.pdf by VELLORE INSTITUTE TECHNOLOGY user on 25 October 2026

https://doi.org/10.1021/acsnano.1c04464
ACS Nano 2021, 15, 19054−19069